

# BioSpecInfo

A guide for the people who use it — BioSpecInfo

---

Samuele Pio Provenzano

Versione **bsi-v187** · commit [7d4d213](#)

Generato il 2026-10-07

Documento generato automaticamente dal sorgente Markdown versionato nel repository. La versione e il commit stampati in ogni piè di pagina identificano esattamente lo stato del software descritto.

# A guide for the people who use it — BioSpecInfo

---

\_This guide is written for people who study, not for people who program. You do not need to know anything about computing.\_

---

## What it is

---

BioSpecInfo is an application for studying chemistry, biochemistry, pharmacology and astrochemistry. It opens like a website, but you can **install** it on your phone or computer and it works **without internet** too.

Three things to know straight away:

- **No sign-up needed.** No account, no email, no password.
- **It costs nothing.**
- **Your data stay yours.** Notes, files and conversations never leave your

device: there is no server they could reach.

👉 [samupropio1-ship-it.github.io/BioSpecInfo-v11](https://samupropio1-ship-it.github.io/BioSpecInfo-v11)

---

## Installing it (optional, but worth it)

---

Installing gives you an icon on your screen, full-screen opening and offline operation.

| Device           | How to do it                                                |
|------------------|-------------------------------------------------------------|
| Android / Chrome | : menu → <i>Install app</i> (or <i>Add to Home screen</i> ) |
| iPhone / iPad    | Share button → <i>Add to Home Screen</i>                    |
| Computer         | Install icon in the address bar, on the right               |

After installing, open the app once with the network on: it will download everything it needs. From then on it works with no signal too.

---

## Finding your way around

---

| Element                                                                                           | Where it is    | What it is for                                        |
|---------------------------------------------------------------------------------------------------|----------------|-------------------------------------------------------|
|  <b>Search</b> | At the top     | Searches the whole app. Shortcut: <code>Ctrl+K</code> |
| <b>Section menu</b>                                                                               | Navigation bar | The 92 sections, grouped by subject                   |
|                | Bottom right   | Tools, settings and updates                           |
| <b>Spectra</b>                                                                                    | Bottom right   | The assistant that answers questions                  |

---

# The things you will do most often

## Analysing a molecule

1. Open the **Spectroscopy centre**
2. Type the molecule into the box. You can use:
  - the **name** — aspirin, caffeine
  - the **SMILES formula** — CC(=O)Oc1ccccc1C(=O)O
  - or **draw it** with the editor
1. Press Enter

You get the 2D and 3D structure, the functional groups, the molecular weight, the predicted spectra (IR, NMR, mass, UV-Vis) and the pharmacological properties.

**If you have no network:** searching by *name* does not work, because it has to consult an external database. The SMILES formula, on the other hand, always works.

## Reading an IR spectrum

The spectrum is drawn as on a real instrument: the scale at the bottom runs from 4000 to 400  $\text{cm}^{-1}$  (**from left to right the numbers go down**) and the bands point **downwards**.

The coloured area below 1500  $\text{cm}^{-1}$  is the **fingerprint region**: that is where the specific molecule is recognised, not just its class. Below the chart you will find the list of bands with their assignment.

The spectra are **predicted** from the structure, not measured in a laboratory. They are there to help you recognise functional groups and to understand the link between structure and spectrum. For experimental data, the app sends you to the official databases.

## Reading a spectrum you measured yourself

The **Spectrum reader** section does the opposite of the one above: it does not predict a spectrum from a structure, it reads one you already have.

1. Drag the file into the box, or paste its text
2. Press **Read**

**JCAMP-DX** files (`.jdx`, `.dx`), which most spectrophotometers produce, work — including the compressed form — and so do two columns of numbers separated by spaces or commas, which is what a spreadsheet produces. If you have no file at hand, the three buttons **IR example**, **MS example** and **NMR example** load a sample spectrum.

Below the chart you will find the peaks found with their signal-to-noise ratio, and for each band **all** compatible assignments. If you read two rows at 1715  $\text{cm}^{-1}$ , that is not a defect: at that wavenumber a ketone and a carboxylic acid are both possible, and naming only one

would be inventing.

The reader **does not tell you which molecule it is**. A set of bands is compatible with many different molecules: the step from spectrum to structure is yours to take, with everything else you know about the sample.

## Choosing the language

The **Languages** section holds **fourteen** languages, with a search field to find them quickly: Italian, English, Spanish, French, German, Portuguese, Dutch, Polish, Romanian, Greek, Russian, Chinese, Japanese and Arabic. Choosing Arabic also flips the page's **direction**, right to left.

Next to it is **Chemical languages**, which is a different thing: it converts a molecule among the twenty-five notations chemistry writes it in — SMILES, InChI, SMARTS, molfile, XYZ, PDB, CML, the fingerprints — and, for a declared class of simple molecules, writes its **IUPAC name**. Outside that class it says so, rather than attempting a wrong name.

## Studying with cards

1. Open the study section
2. Generate a deck of cards
3. For each card, after answering, say how well you remembered it

The app works out **when** to show it to you again: the things you know well come back rarely, the uncertain ones come back soon. When you reopen, you only find the cards that are due.

## Looking up a drug

Open **Pharmacology**. For each of the 263 drugs you will find the structure, molecular weight, mechanism of action, indications, adverse effects and class.

 **For teaching purposes only.** This information does not replace the judgement of a doctor or a pharmacist, and is not there to decide a therapy.

---

## Spectra, the assistant

Spectra answers chemistry questions and can use the app's own tools: calculate, generate spectra, look up constants, balance reactions.

## Getting it working

Spectra relies on an external artificial-intelligence service, and you need a free **key**. Two routes:

**The simplest — a key of your own**

1. Go to [console.groq.com](https://console.groq.com) and create a key

(free, it takes two minutes)

1. Open Spectra, paste the key into the  box

2. Done

### The definitive one — a proxy

If services often fail to answer for you, or you would rather not keep the key on your phone, you can publish a *proxy*: a small free program that keeps the key safe and makes even the services that are blocked from the browser work.

In the assistant's **Proxy** box you will find "**How to set it up, in 4 steps**", with the commands ready to copy.

### When Spectra does not answer

Almost always it is not the app's fault but the external service's. Press  **Test** in Spectra's panel: in a few seconds it tells you **which services your device can actually reach** — and you need no key to find out.

If a service stops answering, Spectra remembers it for 24 hours and moves it to the bottom of the list, under "*Did not answer from this device*". It does not delete it: tomorrow it might work again.

---

## The File Manager

A personal archive for photos, documents and HTML pages, protected by a password.

 **Something you need to know.** This protection is a **deterrent**, not real security: whoever knows where to look can get around it. Do not put confidential documents in it.

The files stay on your device. **There is no copy anywhere else:** if you clear your browser data or change phone, they are gone. Use the export for the things you care about.

---

## Frequently asked questions

**Do I have to pay for anything?** No. The app is free. If you use Spectra, the AI service's free key has daily limits, but nothing is paid.

**Does it really work without internet?** Yes, after the first opening. The only things left out are those that by their nature have to ask someone else: searching molecules by name, 3D structures downloaded on the spot, and Spectra.

**Are my notes saved somewhere?** Only on your device. There is no server: nobody can read them, but nobody can recover them for you either if you lose them.

**How do I delete my data?** ✨ menu → Settings → delete data. You can choose **what**: only the keys, only the conversations, only the study data, or everything.

**I updated but I do not see anything new.** ✨ menu → **Updates** → *Check*. There you can also see which version you are running.

**Can I trust the spectra for a report?** For recognising functional groups and understanding the structure-spectrum relationship, yes. For a figure to be cited as experimental, no: they are predictions. The app sends you to the official databases (NIST, SDBS) with direct links.

**I found an error in a chemical datum.** Report it: open an *issue* on the [GitHub repository](#). The data go through automatic checks, but no check finds everything.

---

## If something goes wrong

| Problem                       | What to try                                                         |
|-------------------------------|---------------------------------------------------------------------|
| The app does not open         | Reload it. If it persists, ✨ menu → Updates → Check                 |
| I do not see the new features | The version is in the ✨ menu → Updates                              |
| Spectra does not answer       | The 🐛 <b>Test</b> button: it tells you which services you reach     |
| "Out of space"                | ✨ menu → Settings → delete the old conversations                    |
| A section stays empty         | Check the network: some sections load data from the internet        |
| The charts are blurred        | Reload the page; the charts are redrawn at your screen's resolution |

---

## Contacts

**Author:** Samuele Pio Provenzano **Repository and reports:** [github.com/samupropio1-ship-it/BioSpecInfo-v11](https://github.com/samupropio1-ship-it/BioSpecInfo-v11)

---

\_Guide updated to version `bsi-v187` . \_